

Exploración Semántica No Determinista y Concurrente ($S_1 \parallel S_2$)

Rama 1 de 3 (SOS)

$$\mathcal{T}_1 = \frac{\langle x := 1, \emptyset \rangle \Rightarrow \sigma_1 [\text{ass}_{\text{sos}}]}{\langle S_1, \emptyset \rangle \Rightarrow \langle (x := 2 ; x := x + 2), \sigma_1 \rangle} [\text{par}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_2 = \frac{\langle x := 2, \sigma_1 \rangle \Rightarrow \sigma_2 [\text{ass}_{\text{sos}}]}{\langle (x := 2 ; x := x + 2), \sigma_1 \rangle \Rightarrow \langle x := x + 2, \sigma_2 \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_3 = \langle x := x + 2, \sigma_2 \rangle \Rightarrow \sigma_3 [\text{ass}_{\text{sos}}]$$

Sentencias Abreviadas

$$S_1 = x := 1 \parallel (x := 2 ; x := x + 2)$$

Estados Registrados

$$\sigma_1 = \{x \mapsto 1\}$$

$$\sigma_2 = \{x \mapsto 2\}$$

$$\sigma_3 = \{x \mapsto 4\}$$

Rama 2 de 3 (SOS)

$$\mathcal{T}_1 = \frac{\langle x := 2, \emptyset \rangle \Rightarrow \sigma_1 [\text{ass}_{\text{sos}}]}{\langle (x := 2 ; x := x + 2), \emptyset \rangle \Rightarrow \langle x := x + 2, \sigma_1 \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_2 = \frac{\mathcal{T}_1}{\langle S_1, \emptyset \rangle \Rightarrow \langle x := 1 \parallel x := x + 2, \sigma_1 \rangle} [\text{par}_{\text{sos}}^3]$$

$\Downarrow$

$$\mathcal{T}_3 = \frac{\langle x := 1, \sigma_1 \rangle \Rightarrow \sigma_2 [\text{ass}_{\text{sos}}]}{\langle x := 1 \parallel x := x + 2, \sigma_1 \rangle \Rightarrow \langle x := x + 2, \sigma_2 \rangle} [\text{par}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_4 = \langle x := x + 2, \sigma_2 \rangle \Rightarrow \sigma_3 [\text{ass}_{\text{sos}}]$$

Sentencias Abreviadas

$$S_1 = x := 1 \parallel (x := 2 ; x := x + 2)$$

Estados Registrados

$$\sigma_1 = \{x \mapsto 2\}$$

$$\sigma_2 = \{x \mapsto 1\}$$

$$\sigma_3 = \{x \mapsto 3\}$$

Rama 3 de 3 (SOS)

$$\mathcal{T}_1 = \frac{\langle x := 2, \emptyset \rangle \Rightarrow \sigma_1 [\text{ass}_{\text{sos}}]}{\langle (x := 2 ; x := x + 2), \emptyset \rangle \Rightarrow \langle x := x + 2, \sigma_1 \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_2 = \frac{\mathcal{T}_1}{\langle S_1, \emptyset \rangle \Rightarrow \langle x := 1 \parallel x := x + 2, \sigma_1 \rangle} [\text{par}_{\text{sos}}^3]$$

$\Downarrow$

$$\mathcal{T}_3 = \frac{\langle x := x + 2, \sigma_1 \rangle \Rightarrow \sigma_2 [\text{ass}_{\text{sos}}]}{\langle x := 1 \parallel x := x + 2, \sigma_1 \rangle \Rightarrow \langle x := 1, \sigma_2 \rangle} [\text{par}_{\text{sos}}^4]$$

$\Downarrow$

$$\mathcal{T}_4 = \langle x := 1, \sigma_2 \rangle \Rightarrow \sigma_3 [\text{ass}_{\text{sos}}]$$

Sentencias Abreviadas

$$S_1 = x := 1 \parallel (x := 2 ; x := x + 2)$$

Estados Registrados

$$\sigma_1 = \{x \mapsto 2\}$$

$$\sigma_2 = \{x \mapsto 4\}$$

$$\sigma_3 = \{x \mapsto 1\}$$